Exercises: Influence Lines

Lab S2 – Cantilever (Gerber) Bridge Model

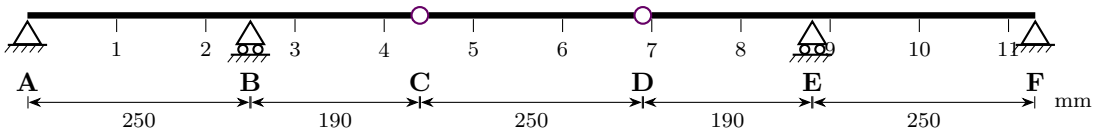


Name: _____ Matric No.: _____ Group: _____

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Before you start

All questions refer to the laboratory model (geometry below; $\eta_R(x)$ denotes the influence line of R for a unit load at x measured from A, support reactions positive upwards, hinge forces positive when pressing down on the cantilever tips). Work on paper first; the *Influence Line Explorer* (scan the code) lets you check Questions 1, 3, 4 and 10 afterwards. Total: 60 marks (plus 4 bonus).



Part A – Reading and deriving influence lines

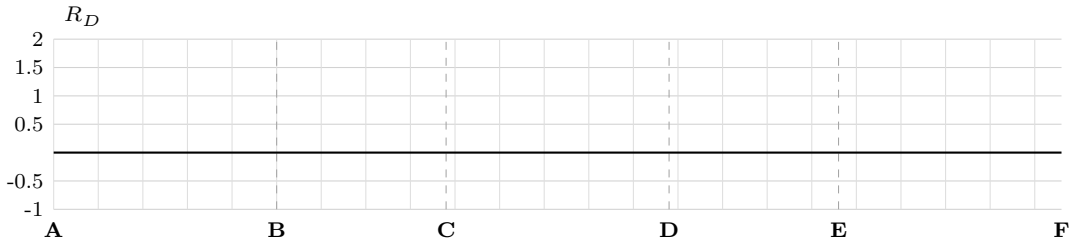
Question 1. (6 marks) Using the formulae derived in the lecture notes, find the ordinates

(a) η_A and η_B for a unit load at $x = 150$, 350 and 560 mm;

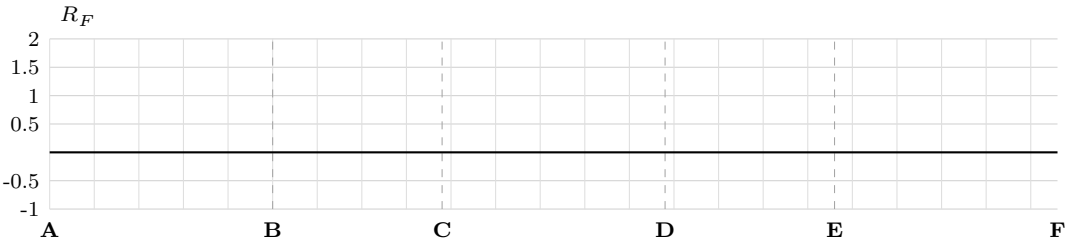
(b) η_E and η_F for a unit load at $x = 760$ and 1000 mm.

Check each pair with $\eta_A + \eta_B + \eta_E + \eta_F = 1$.

Question 2. (8 marks) Starting from the free-body diagram of the suspended span and of beam DEF, derive the influence lines of the hinge force R_D and of the reaction R_F for $0 \le x \le 1130$ mm. Give the expressions for each range of x and sketch both lines to scale on the grids below, marking the ordinates at C, D, E and F.



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Part B – Using influence lines

Question 3. (8 marks) A model vehicle has two axles 150 mm apart carrying 4 kg (leading) and 2 kg (trailing). It moves from A to F.

(a) Explain why the largest value of R_B must occur with one of the two axles standing on a vertex of η_B.

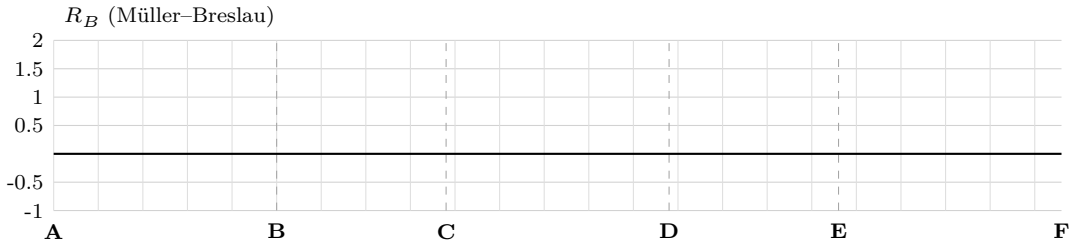
(b) Find the position of the vehicle that gives the largest R_B and the value of R_B (in kgf).

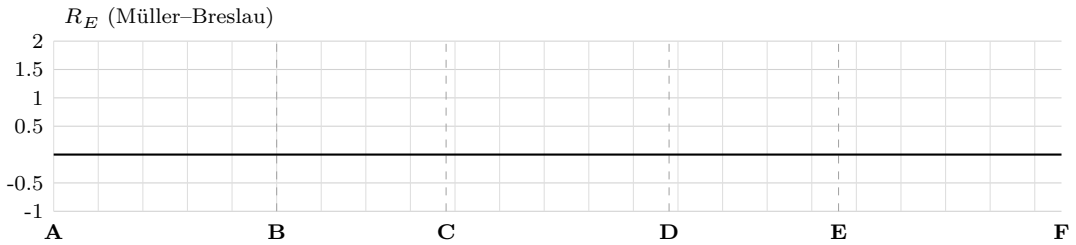
(c) Find the position that gives the most negative R_A and the value of R_A.

Question 4. (8 marks) A uniformly distributed load $w = 0.02$ kg/mm is placed (a) over AB only, (b) over the suspended span CD only, (c) over the whole model. For each case find R_A, R_B and the hinge force R_C from the areas under the influence lines. For case (c) check that the four support reactions add up to the total load.

Question 5. (6 marks) Use the Müller–Breslau principle to sketch the influence lines of R_B and R_E (release the support, impose a unit displacement, draw the deflected shape). Mark the ordinates at A to F. Then, using η_F from Question 2 and the symmetry of the model, state which supports can experience uplift and for which positions of the load.

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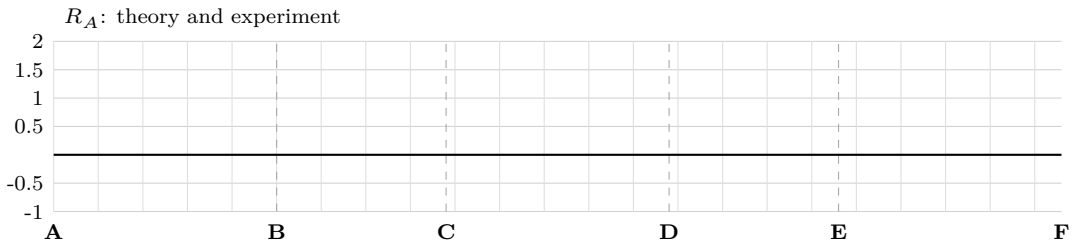




Part C – The experiment

Question 6. (8 marks) The readings below are *illustrative* dial-gauge readings for gauge A with the 2.5 kg mass at each halt (Test 1); one division = 0.05 kgf. Complete the table: convert each reading to the experimental ordinate η_{test}, write the theoretical ordinate, and the difference. Plot both lines on the grid and comment on the agreement in two or three sentences.

Halt	x (mm)	Reading (div)	η_{test}	η_{theory}	Difference
A	0	51			
1	100	29			
2	200	10			
B	250	1			
3	300	-11			
4	400	-28			
C	440	-39			
5	500	-29			
6	600	-13			
D	690	-1			
7	700	0			
8	800	0			
E	880	1			
9	900	-1			
10	1000	0			
11	1100	1			
F	1130	0			



Question 7. (6 marks)

(a) The mass can be placed within ±5 mm of a halt and the gauge is read to ±1 division. Estimate the resulting uncertainty of an experimental ordinate of η_A for a halt on AB and for a halt on CD.

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(b) Taking the combined uncertainty as ± 0.03 per unit load in round numbers, for which halts with a non-zero theoretical ordinate could this uncertainty alone give a percentage difference larger than 10 %? Hence explain why the report should compare ordinates by absolute difference.

Question 8. (4 marks) With the 2.5 kg mass at halt 4, the four support gauges read $a = -28$, $b = 77$, $e = 0$ and $f = 0$ divisions. Check the equilibrium of the model and interpret the result.

Part D – Discussion

Question 9. (6 marks) Answer briefly.

(a) Why does the Gerber arrangement (two hinges in the central span) make the bridge statically determinate, and what does this imply for the shape of every influence line?

(b) If the hinges at C and D were replaced by rigid welds, so that ABCDEF became a continuous beam on four supports, how would the influence line of R_A change? Sketch its shape with the Müller–Breslau principle.

(c) Give one practical reason why a designer must know that R_A can be negative.

Question 10 (challenge, bonus). (4 marks) The Test 2 pair now crosses the bridge from F towards A, so the 2.5 kg mass is 100 mm to the *left* of the 5 kg mass. Find the most negative R_A, the largest hinge force R_C and the position of the 5 kg mass that gives it. Compare with Worked example 1 of the lecture notes.

End of exercises. Marks: Part A 14, Part B 22, Part C 18, Part D 6 (total 60), plus 4 bonus marks for Question 10.

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